SESSION 6

UML Class Diagram Design

AI Career Guidance & Skill Recommendation System

Project	AI Career Guidance & Skill Recommendation System
Deliverable	Class Diagram
Industry Mapping	Object-Oriented System Design
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Repository	[Add GitHub repository link]

Topics Covered in This Session

- Object-oriented analysis
- Class identification, attributes, methods, relationships
- UML class diagram modelling

1. Object-Oriented Analysis

Candidate classes were found by underlining **nouns** in the SRS (user, resume, skill, career, recommendation, gap, roadmap, course, chat) and assigning **verbs** as operations (upload, parse, match, generate, recommend). Classes without data or behaviour (for example "page" or "button") were dropped as UI details.

Noun in SRS	Decision	Reason
Student, Admin	Classes (subclasses of User)	Different permissions and behaviour, shared identity data
Resume	Class	Holds file, text and parsed profile
Skill, Career, Course	Classes	Core catalogue entities managed by admin
Recommendation, SkillGapReport, Roadmap, RoadmapStep	Classes	Results generated per student
ChatSession, ChatMessage	Classes	Conversation history
Claude, Analytics	Service classes	Encapsulate external AI and Pandas analytics
Dashboard, Page, Button	Rejected	User interface elements, not domain classes

2. Class Diagram

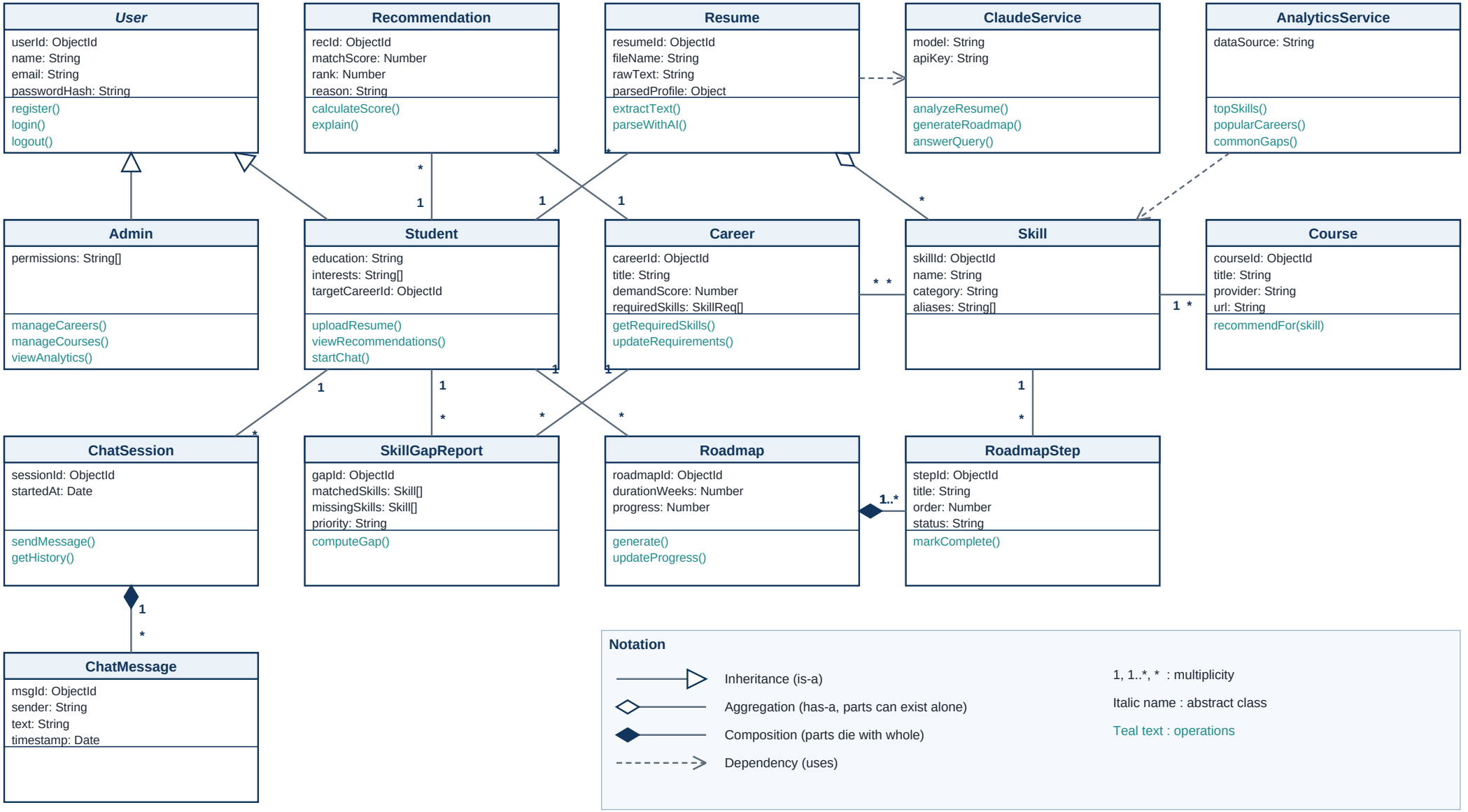

Figure 1: UML class diagram of the AI Career Guidance & Skill Recommendation System

3. Class Descriptions

Class	Responsibility	Key attributes	Key methods
User (abstract)	Common identity and authentication	userId, name, email, passwordHash	register(), login(), logout()
Student	End user who seeks guidance	education, interests, targetCareerId	uploadResume(), viewRecommendations(), startChat()
Admin	Maintains catalogue and users	permissions	manageCareers(), manageCourses(), viewAnalytics()
Resume	Stores uploaded file data and AI-parsed profile	fileName, rawText, parsedProfile	extractText(), parseWithAI()
Skill	A single skill used across careers and courses	name, category, aliases	(data class)
Career	A job role with weighted required skills	title, demandScore, requiredSkills	getRequiredSkills(), updateRequirements()
Recommendation	Ranked career result for a student	matchScore, rank, reason	calculateScore(), explain()
SkillGapReport	Matched and missing skills for a target career	matchedSkills, missingSkills, priority	computeGap()
Roadmap	Ordered learning plan with progress	durationWeeks, progress	generate(), updateProgress()
RoadmapStep	One learning step tied to a skill	title, order, status	markComplete()
Course	Learning resource linked to a skill	title, provider, url	recommendFor(skill)
ChatSession / ChatMessage	Conversation container and its messages	sessionId; sender, text, timestamp	sendMessage(), getHistory()
ClaudeService	Single gateway to the Claude API	model, apiKey	analyzeResume(), generateRoadmap(), answerQuery()
AnalyticsService	Aggregates data for the dashboard	dataSource	topSkills(), popularCareers(), commonGaps()

4. Relationships

Relationship	Type	Multiplicity	Meaning
Student, Admin to User	Inheritance	-	Both are kinds of User
Student to Resume	Association	1 to *	A student uploads many resume versions
Resume to Skill	Aggregation	1 to *	A resume contains skills that also exist in the catalogue
Student to Recommendation	Association	1 to *	A student receives many recommendation results
Recommendation to Career	Association	* to 1	Each recommendation refers to one career
Career to Skill	Association	* to *	Careers require many skills; skills serve many careers
Student to SkillGapReport; Report to Career	Association	1 to *; * to 1	Gap reports belong to a student and target a career
Student to Roadmap	Association	1 to *	A student can have a roadmap per target career

Relationship	Type	Multiplicity	Meaning
Roadmap to RoadmapStep	Composition	1 to 1..*	Steps cannot exist without their roadmap
RoadmapStep to Skill; Course to Skill	Association	* to 1	Steps and courses are tied to one skill
Student to ChatSession; Session to Message	Association; Composition	1 to *	Messages belong to a session
Resume to ClaudeService; AnalyticsService to Skill	Dependency	-	Uses an external or helper service

Design notes

- Services (ClaudeService, AnalyticsService) keep AI and analytics logic out of domain classes, supporting low coupling.
- Match score logic sits in Recommendation.calculateScore() using Career.requiredSkills weights, as defined in FR-12.
- Class attributes map directly to the MongoDB collections listed in the SRS.